Royal Devon University Healthcare NHS Foundation Trust

Critical Appraisal: Study Designs

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Learning Objectives

This session will help you:

- Understand why certain types of research are more ‘powerful’ than others
- Identify the key features of common study designs
- Assess the appropriate study design for a clinical/research question
- Distinguish between primary and secondary research

Levels of Evidence

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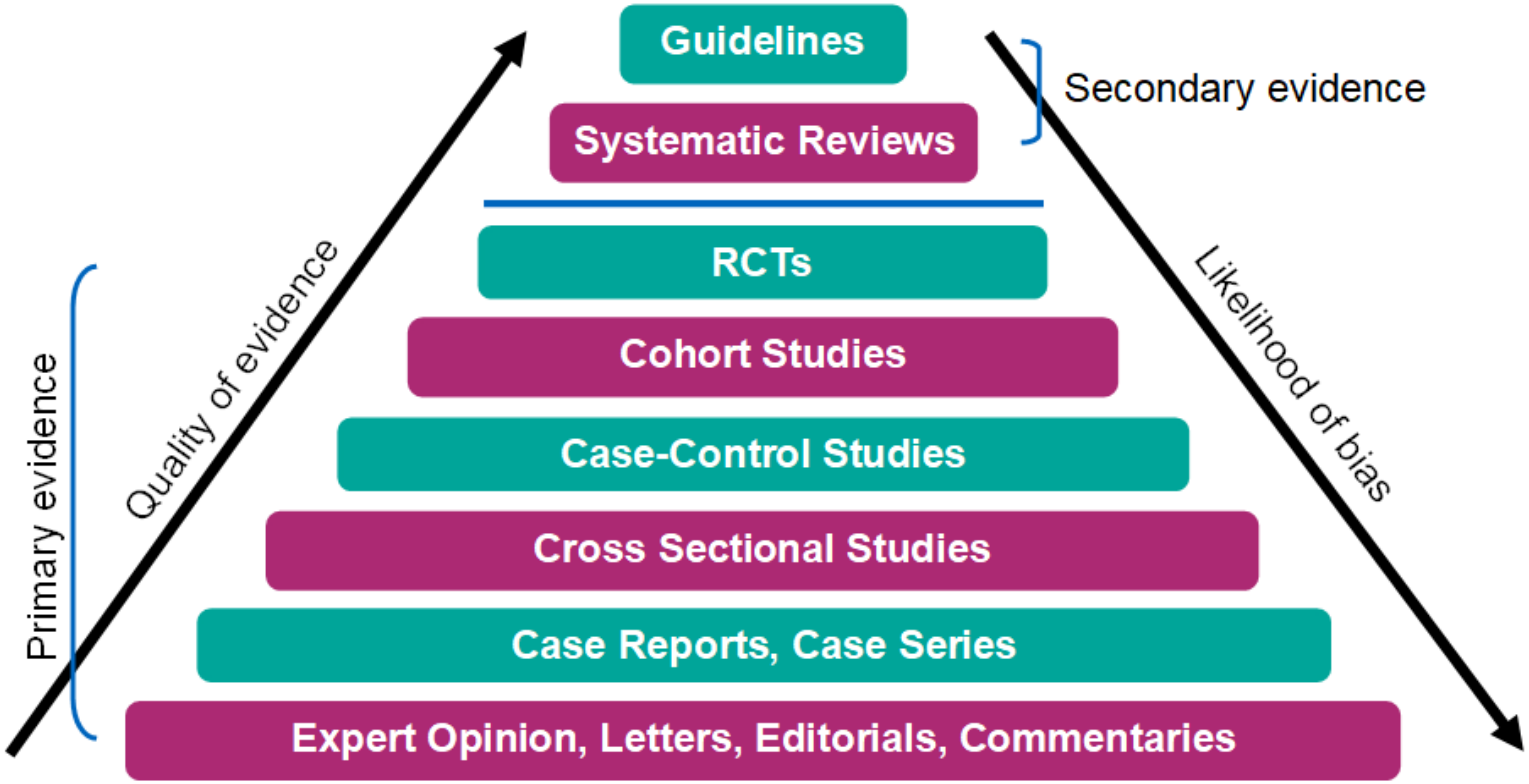

Definition of Bias

"A bias is a systematic error, or deviation from the truth, in results or inferences. Biases can operate in either direction; different biases can lead to underestimation or overestimation of the true intervention effect".

Cochrane Handbook

- Selection Bias
- Attrition Bias
- Recall Bias
- Observation Bias

Levels of Evidence

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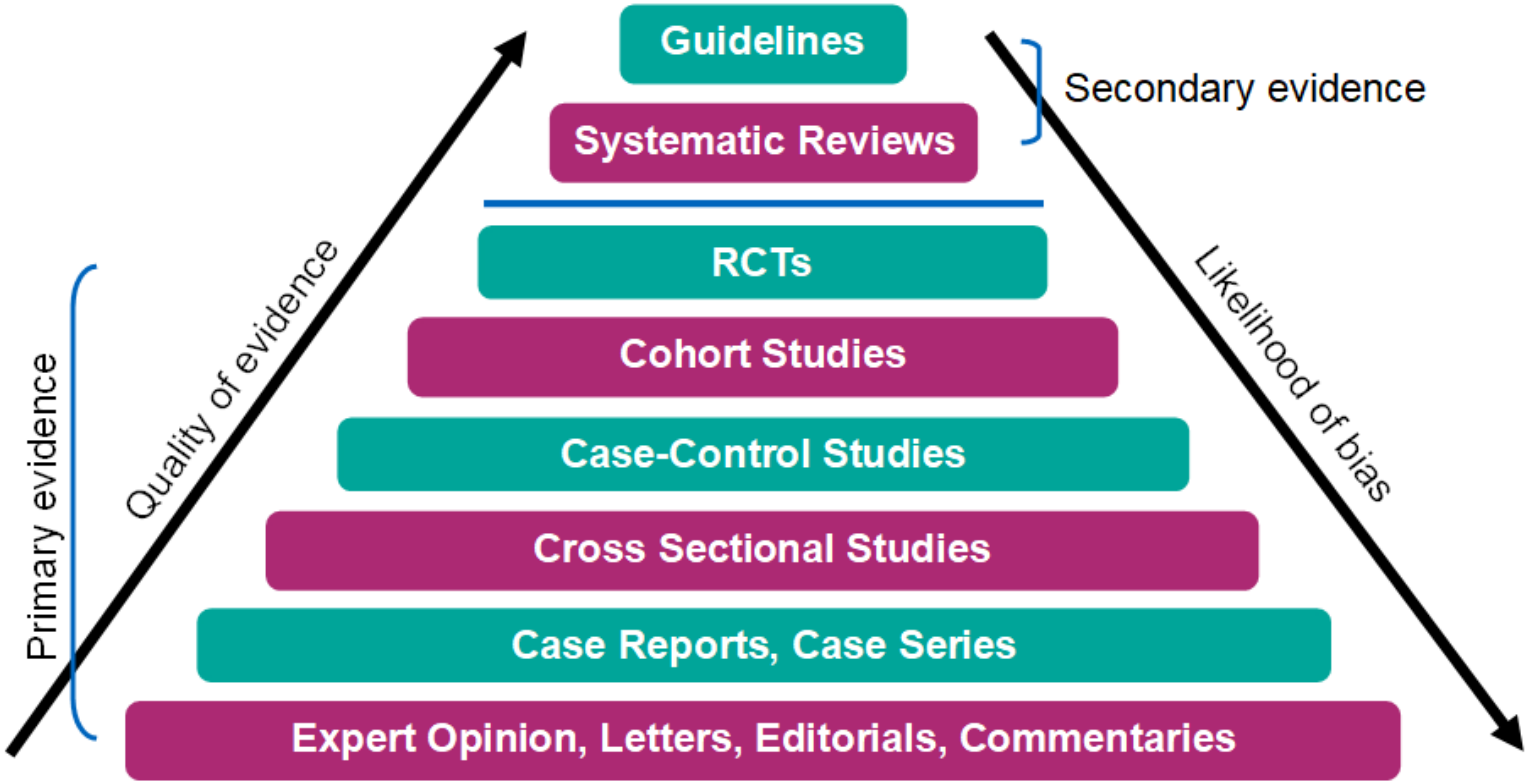

Case Report

Observational, descriptive, narrative

Key features:

- Short report detailing case of particular patient
- Can sometimes be accompanied by the short literature review situating the case within broader context

Advantages

- Can identify new/emerging/rare conditions
- Used to generate a hypothesis
- Highlight unusual things

Disadvantages

- Single case
- Anecdotal
- Generally not repeatable
- Chance/bias

Case Series

Observational, descriptive, narrative

Key features:

- Narrative analysis of set of rare cases

Advantages

- Can identify new emerging trends/adverse effects
- Indicates areas for further research

Disadvantages

- Small samples

Cross-Sectional Studies

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Observational, analytical

Key features:

- Often questionnaire based
- No follow-up

Advantages

- Can recruit large numbers
- Can indication an association

Disadvantages

- No follow-up = no information on temporal sequence
- Cannot determine cause and effect
- Depend on recollection

Case-Control Studies

Patients (cases) who have developed a condition are identified and their exposure to suspected aetiological factors is compared to a group (control) who have not developed that condition.

Key features:

- Case group and control group often matched by age, sex etc. to help eliminate confounding variables
- Strict case definition – severity, location of presentation
- Sometimes utilise two sets of controls – greater statistical confidence
- Generally questionnaire based but can use historical patient records

Advantages

- Can be used to study diseases with lower incidence

Disadvantages

- Control can be difficult to establish – confounding factors Often depends on recall
- Case group more motivated to find cause – controls have no special interest

Cohort Studies

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Observational, analytical. Can be prospective or retrospective

Key features:

- Longitudinal – cohorts will be followed up over time
- Can be one cohort, but more powerful with two

Advantages

- Can identify aetiology/prognosis
- Can estimate disease incidence rates
- Can assess temporality

Disadvantages

- Long studies – can take time to develop outcome.
- Subjects drop out = attrition bias

Randomised Controlled Trials

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Experimental, prospective

Key features:

- Randomisation
- Blinding (single/double/triple)
- Statistical analysis
- Crossover/wash out period

Advantages

- Can recruit large numbers = more power
- Employs a number of techniques throughout to minimise bias
- Reliable measure of efficacy
- Can be used for meta-analysis

Disadvantages

- Expensive – look for who is paying for them
- Bias still possible throughout – drop outs, selection, reporting

Systematic Reviews

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Very high secondary evidence, compiled by teams of people e.g. Cochrane

Key features:

- Search methodology
- Selection criteria
- Meta-analysis – Forest plot

Advantages

- Stronger than individual primary studies
- Offer a precis of the best primary studies – save time
- Minimise bias with strict inclusion/exclusion criteria e.g. number of subjects

Disadvantages

- Some are better than others…

Forest Plot

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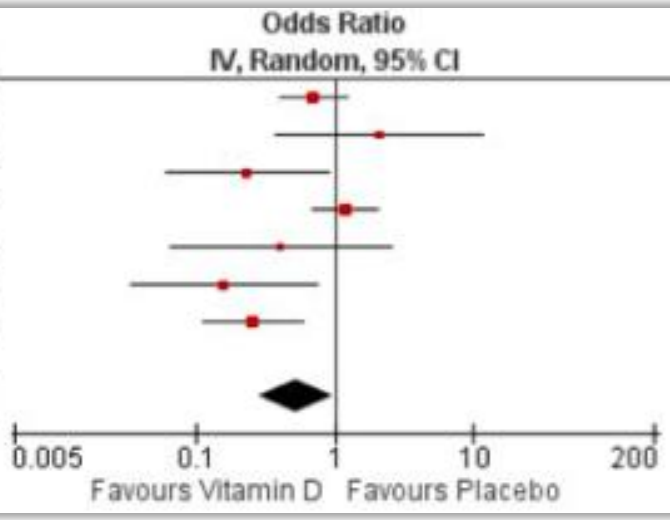
Study or Subgroup	log[Odds Ratio]	SE	Vitamin D Total	Placebo Total	Weight	Odds Ratio IV, Random, 95% CI
Castro 2014	-0.3567	0.2855	201	207	21.4%	0.70 [0.40, 1.22]
Jensen 2016	0.7419	0.8715	11	11	8.9%	2.10 [0.38, 11.59]
Majak 2011	-1.4424	0.684	24	24	11.9%	0.24 [0.06, 0.90]
Martineau 2015	0.1823	0.275	108	114	21.7%	1.20 [0.70, 2.06]
Tachimoto 2016	-0.891	0.9401	54	35	8.1%	0.41 [0.06, 2.59]
Urashima 2010	-1.8334	0.7906	51	59	10.1%	0.16 [0.03, 0.75]
Yadav 2014	-1.3499	0.4272	50	50	17.8%	0.26 [0.11, 0.60]
Total (95% CI)			**499**	**500**	**100.0%**	**0.53 [0.28, 0.99]**

Heterogeneity: Tau² = 0.40; Chi² = 17.04, df = 6 (P = 0.009); I² = 65%

Test for overall effect: Z = 1.98 (P = 0.05)



Thank You for Listening!